6.4 Nuclear and particle physics

This section provides knowledge and understanding of the atom, nucleus, fundamental particles, radioactivity, fission and fusion.

Nuclear power stations provide a significant fraction of the energy needs of many countries. They are expensive; governments have to make difficult decisions when building new ones. The building of nuclear power stations can be used to evaluate the benefits and risks to society (HSW9). Ethical, environmental and decision making issues may also be discussed (HSW10 and HSW12). The development of the atomic model also addresses issues of scientific development and validation (HSW7, 11).

6.4.1 The nuclear atom

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	alpha-particle scattering experiment; evidence of a small charged nucleus	HSW7
(b)	simple nuclear model of the atom; protons, neutrons and electrons	
(c)	relative sizes of atom and nucleus	*M0.4, M1.4*
(d)	proton number; nucleon number; isotopes; notation ${}^{A}_{Z}X$ for the representation of nuclei	
(e)	strong nuclear force; short-range nature of the force; attractive to about 3 fm and repulsive below about 0.5 fm	1 fm = 10^{-15} m
(f)	radius of nuclei; $R = r_0 A^{1/3}$ where r_0 is a constant and A is the nucleon number	
(g)	mean densities of atoms and nuclei.	HSW7